

Philosophy of Mathematics Essay

WHAT PROBLEM DOES THE EXISTENCE OF MATHEMATICAL
STRUCTURES WITHIN INDISCERNIBLE ELEMENTS POSE FOR
ANTE REM STRUCTURALISM?

nº 163986

School of Arts and Humanities (FLUL), University of Lisbon

Philosophy of Mathematics

2nd Semester 2025/26

Professor Bruno Jacinto

Contents

1	Introduction	3
2	The Indiscernibility Objection	4
3	The graph case and the Leitgeb-Ladyman answer	5
4	Why is this better than a compromise?	6
5	Isn't this just Old News?	7
6	Conclusion	8
7	References	9

1 Introduction

Ante rem structuralism is the thesis that mathematical objects are positions in structures that themselves exist independently of the systems that instantiate or exemplify them [7, 9]. However, in that case, the existence of structurally indiscernible positions seems to pose a relevant issue: if no structural property distinguishes them, are they the same position, or are they distinguishable by some non-structural property?

MacBride formulates this issue in a dilemma: either the structuralist accepts the "bad news", meaning, the indiscernible positions collapse into one; or she accepts the "old news", that the difference between positions relies on extra-structural properties, namely irreducible identity facts, and the view becomes a version of Platonism [6].

I defend that the best way for the *ante rem* structuralist to reply to the dilemma is neither to accept any of its options, nor to adopt Button's hybrid solution. The most promising answer is Leitgeb and Ladyman's (L&L): the difference between positions can be internal to the structure without being a qualitative local property or a haecceity. This strategy addresses the assumption on which the problem depends: that numerical difference requires qualitative individuation. By denying this, the difference between positions can belong to the global specification of the structure itself, without relying on haecceitism and preserving the structuralist stance.

The argument proceeds in four steps. Firstly, I reconstruct the indiscernibility objection and explain why it seems to threaten *ante rem* structuralism. Second, I distinguish this case from stronger cases of indistinguishability, mainly the unlabelled two-node edgeless graph discussed by L&L. Third, I compare L&L's response with Button's hybrid solution, arguing that Button avoids the hardest cases by treating them eliminativistically, while L&L answer them directly. Lastly, I consider the objection that L&L merely reintroduce platonism, and argue that this objection fails because their view does not appeal to haecceities: the distinctness of the nodes is part of the structure itself.

2 The Indiscernibility Objection

The indiscernibility objection, as presented by Keränen [3] can be reconstructed as follows:

1. Ante rem structuralism claims that: (a) Mathematical objects are positions in structures; (b) Positions have no intrinsic non-structural nature.
2. Therefore, if two positions are distinct, their distinctness must be structurally explained.
3. Automorphic positions share all invariant structural properties.
4. So ante rem structuralism is unable to explain their distinctness.

In fact, in the complex number field, complex conjugation preserves the structure but maps i onto $-i$,

$$(\mathbb{C}, +, \cdot, 0, 1), \quad z \mapsto \bar{z}, \quad i \mapsto -i.$$

In this sense, the pure structure doesn't privilege one root over the other, exemplifying the kind of non-trivial automorphism Keränen refers to. MacBride makes this even more evident, by constructing a dilemma:

MacBride's Dilemma (MD): (1) Bad News: if indiscernibility means identity then positions mathematics treats as distinct collapse, i.e, $i = -i$; or (2) Old News: if $i \neq -i$ simply in a primitive way, then the view collapses to Platonism.

Indeed, if $i = -i$, then we arrive at an unacceptable conclusion. And if identity facts are understood as haecceities, then structuralists must abandon their own motivation that that mathematical objects have no nature beyond their structural role. So if taking the routes (1) or (2) would almost certainly make structuralism indefensible, the possible routes left are the following: (3) dissolving the dilemma, by showing it is possible and consistent with the structuralist thesis to deny both (1) and (2); or (4) accepting the issues Keränen raises and restructure the thesis, forming a hybrid position.

Shapiro [10] takes route (3) by appealing to a semantic dimension where i functions as a parameter.¹ Still, while this constitutes a strong semantic reply, it does not answer the ontological question of how there can exist different genuine positions when there's no structural distinction to them.

Moreover, while i and $-i$ are automorphic in the complex field, they are weakly discernible, because they're in the relation of being inverse, non-null, additives of one another [4, 2].²

However there are mathematical structures with positions that are not even weakly discernible. A strong reply will use these structures to argue for the denial of (1) without falling back to (2).

3 The graph case and the Leitgeb-Ladyman answer

Consider the graph³:

$$G' = (V, E), \quad V = \{u, v\}, \quad E = \emptyset.$$

G' is an unlabelled graph with two isolated nodes u and v , swapping them preserves the graph, so there's no structural relation that distinguishes them locally. In this sense, unlike with the i and $-i$ case, these two positions are not even weakly discernible.

Thus, Button takes route (4) and concludes that structuralism should become hybrid: realist for structures with discernible or weakly discernible positions and eliminative for structures with indistinguishables, such as G' [2].

However, Leitgeb and Ladyman defend that the difference between the positions can be internal to the structure and not explained by something external to it [5]. To describe

¹In this sense, i distinguishes linguistically what the pure structure doesn't metaphysically. And the fact that mathematical language allows for certain useful conventions, doesn't imply that those conventions reveal a metaphysical hidden feature.

²Two objects are weakly discernible when they don't have monadic properties that distinguish them, but they're in a irreflexive relation. A relation is irreflexive when no object is in that relation to itself. For i and $-i$: x and y are non-null and $x + y = 0$

³Example in [5]

a structure is to describe it up to isomorphism. We can express this as:

$$G \cong H \iff \exists f : V_G \rightarrow V_H \text{ bijective such that } \{x, y\} \in E_G \iff \{f(x), f(y)\} \in E_H.$$

Cardinality is invariant under isomorphisms ($G \cong H$ implies $|V_G| = |V_H|$)⁴. Hence, if

$$G' = (\{u, v\}, \emptyset) \quad \text{and} \quad G_1 = (\{w\}, \emptyset),$$

then

$$|V_{G'}| = 2 \neq 1 = |V_{G_1}|,$$

so

$$G' \not\cong G_1,$$

meaning, G' is structurally distinct from the edgeless graph with only one node, even though its two nodes don't have any intrinsically distinct marks.

Consequently, symmetry in structure is not a problem in describing its objects, it's part of what the structure is describing. So, this answer pivots the question: Instead of showing what property distinguishes the node locally, it asks what structure is being specified. The answer is: an edgeless, unlabelled graph of order two.

4 Why is this better than a compromise?

Button is right to insist that the problem isn't solved by answering only to weakly discernible cases. But Button's solution is to restrict structural realism by claiming that we can be realists about structures that have discernible or weakly indiscernible positions, but that we should treat pure indiscernible structures in an eliminativist way, as there is something metaphysically suspicious about positions in these structures [2]. Indeed, this avoids the objection, but only by circumventing it, meaning, we withdraw realism from precisely the cases that make the objection difficult.

⁴Since a direct consequence of isomorphic graphs is that they must have the same number of vertices, but also the same number of edges, same degree sequence, etc.

The cost of this answer is not just a reduced thesis. I argue that the main cost is to concede too much to the critic. We conceive that a structure is ontologically acceptable only if its positions can be qualitatively individuated.

For that, L&L's route is directly opposite: instead of describing indiscernible structures as ontologically suspicious, they refuse the very supposition that makes them suspicious in the first place: that numerical difference demands qualitative individuation. Concretely, they do not evade the hard case by changing the status of G' , but instead show that mathematical practice already treats such graphs as legitimate structures. Consequently, the philosophical demand for a further individuating feature is not imposed by graph theory itself, but by an external metaphysical assumption.

Therefore, the answer is stronger: it is aligned with mathematical practice and it doesn't circumvent MD, but shows that the objection has a hidden assumption.

Hidden assumption: Numerical distinctness requires a qualitative criterion of identity.

5 Isn't this just Old News?

The critic can argue that saying that the graph has order two seems to presuppose what it means to explain: If nothing distinguishes u from v then how can we say that there are two places instead of one?

This objection highlights that L&L don't provide a reductive criterion for identity. They do not explain the difference between the two nodes by pointing to a property that one has and the other lacks. So it can seem that we fall back into (2): there is an intrinsic *thisness* that the node possesses independently of its structural role (haecceity).

However, the critic assumes what L&L specifically aim to reject: that the numerical difference is grounded on a qualitative mark intrinsic to each object. In G' , the structure has two positions because it is a graph of order two, that is part of its global description, not some extra metaphysical postulate.

Moreover, this does not amount to haecceitism, because if this was the case, there

would be some further fact about which node is really u or v , but the structure gives no such fact: permuting the two nodes does not generate a new graph. Their distinctiveness is a structural fact: a symmetrical edgeless graph of order two, distinct from other structures such as the graph with a single node.

6 Conclusion

The indiscernibles objection is serious because it emphasises that structural roles don't always account for qualitative individuation. But this only defeats a reductive version structuralism, not *ante rem* structuralism where identity of places can be part of a global structure description.

For this, L&L's answer is preferable to Button's hybrid solution because it preserves realism about symmetrical structures while rejecting the assumption on which the objection depends: that numerical difference requires qualitative individuation.

Hence, we can conclude that the critic shouldn't ask the structuralist to recover individuality inside mathematical structures in order to argue against this position, but instead ask what model of individuality does mathematics really need. Indiscernible elements don't refute *ante rem* structuralism, only force it to abandon the demand for a reductive criterion of individuation and to understand identity a way more closely in line with mathematical practice.

1471 words

7 References

- [1] Benacerraf, P. (1965). “What Numbers Could Not Be.” *Philosophical Review*, 74(1), 47–73.
- [2] Button, T. (2006). “Realistic Structuralism’s Identity Crisis: A Hybrid Solution.” *Analysis*, 66(3), 216–222.
- [3] Keränen, J. (2001). “The Identity Problem for Realist Structuralism.” *Philosophia Mathematica*, 9(3), 308–330.
- [4] Ladyman, J. (2005). “Mathematical Structuralism and the Identity of Indiscernibles.” *Analysis*, 65(3), 218–221.
- [5] Leitgeb, H. and Ladyman, J. (2008). “Criteria of Identity and Structuralist Ontology.” *Philosophia Mathematica*, 16(3), 388–396.
- [6] MacBride, F. (2005). “Structuralism Reconsidered.” In S. Shapiro (ed.), *The Oxford Handbook of Philosophy of Mathematics and Logic*. Oxford: Oxford University Press, 563–589.
- [7] Reck, E. and Schiemer, G. (2019). “Structuralism in the Philosophy of Mathematics.” *The Stanford Encyclopedia of Philosophy*. Edward N. Zalta and Uri Nodelman (eds.). <https://plato.stanford.edu/entries/structuralism-mathematics/>
- [8] Schiemer, G. and Wigglesworth, J. (2019). “The Structuralist Thesis Reconsidered.” *British Journal for the Philosophy of Science*, 70(4), 1201–1226.
- [9] Shapiro, S. (1997). *Philosophy of Mathematics: Structure and Ontology*. Oxford: Oxford University Press.
- [10] Shapiro, S. (2008). “Identity, Indiscernibility, and Ante Rem Structuralism: The Tale of i and $-i$.” *Philosophia Mathematica*, 16(3), 285–309.